MILITARY SPECIFICATION

COMMANDER'S DISPLAY FOR AN/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Commander's Display for the Sight, Thermal, Tank Mounted, AN/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible for Packaging
MIL-E-49383	- Eyepiece, Biocular, AN/VSG-2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Palletized Type Base
MIL-STD-150	- Photographic Lenses
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment
MS-24535	- Fastener-Container Closure

DRAWINGS

DLSM-B-804913	- Commander's Display, AN/VSG-2
SM-C-805906	- Image Intensifier, 40 Millimeter

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. The Commander's Display for AN/VSG-2, herein known as the commander's display, relays an image of the infrared scene produced by the AN/VSG-2 from the gunner's station to the commander's station.

3.2 Construction. The commander's display shall be constructed in accordance with DLSM-B-804913 and as specified herein.

3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish five (5) commander's displays for first article inspection and approval.

3.4 Parts, materials and processes. Parts, materials and processes shall be as specified herein and as shown on the applicable drawings. Parts, materials and processes not specified shall be selected in accordance with MIL-P-11268.

3.5 Major components. The commander's display consists of the following components and subassemblies and shall be constructed in accordance with applicable drawings, specifications and as specified herein.

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a. Biocular Eyepiece Display - MIL-E-49383
b. Image Intensifier - SM-C-805906
c. Optical Relay Assembly
d. Display Assembly

3.6 Preconditioning. Each commander's display shall be subjected to a 48-hour preconditioning prior to delivery.

3.7 Operation.

3.7.1 Luminance uniformity. With a uniform scene input, the luminance of each display shall not vary more than ±15 percent from the average display luminance over the central 50 percent of the display.

3.7.2 Power. The commander's display power requirements shall not exceed 5 watts.

3.7.3 Optical transfer function (OTF). The modulation transfer function (MTF) shall satisfy the performance characteristics as shown below over the center of the display in the scan direction, with the system's gain set so that the system is operating within its dynamic range and below saturation.

Optical transfer function

f (cycles/milliradian) at input of biocular	MTF
0.06	0.98
0.13	0.92
0.27	0.78
0.36	0.69
0.41	0.64
0.47	0.57
0.55	0.50

3.7.4 Apparent angular magnification. The ratio of the image angular subtense to the corresponding object angular subtense shall not be less than 1.05 times, when viewed through the biocular eyepiece.

3.7.5 Spatial uniformity. The variance in spatial uniformity of the commander's display shall not exceed ±10 percent in the horizontal or vertical directions over the entire display.

3.7.6 Effective focal length (EFL). The effective focal length of the relay system, measured without the image intensifier (I^2) assembly shall be -62, ±3 mm.

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3.7.7 f/number. The f/number shall be f/5.6.

3.7.8 Gain. The commander's display shall have a minimum luminance gain of 20.

3.7.9 Resolution. The visual axial resolution of the commander's display shall be not less than 20 line pairs per millimeter.

3.7.10 Controls.

3.7.10.1 Contrast. A single control shall be provided to simultaneously control the gain of all channels to provide a minimum of 24 dB of attenuation.

3.7.10.2 Polarity switch. A polarity switch shall be provided to reverse the image contrast and cause the display to represent hot or cold targets as either bright or dark. Switching polarity shall not require an adjustment of the contrast or brightness controls.

3.7.10.3 Brightness. A control shall be provided for setting the background level of the image.

3.7.10.4 Commander and gunner. A control shall be provided for the commander that will activate either the commander's or the gunner's brightness, contrast, and polarity controls.

3.7.11 Sealing. The commander's display shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5, ±1 psig for a minimum of 2 hours. The interior atmosphere of the units shall contain dry nitrogen (having a maximum dew point of -32°C at atmospheric pressure), at a pressure of 0.25 to 5.0 psig when delivered.

3.8 Environment.

3.8.1 Low temperature.

3.8.1.1 Operation. The commander's display shall not be damaged (see 6.3) when operated at temperatures to -32°C.

3.8.1.2 Storage. The commander's display shall not be damaged when subjected to storage at temperatures to -57°C.

3.8.2 High temperature.

3.8.2.1 Operation. The commander's display shall not be damaged when operated at temperatures to +52°C.

3.8.2.2 Storage. The commander's display shall not be damaged when stored at temperatures to +71°C.

3.8.3 Temperature shock. The commander's display shall not be damaged when subjected to sudden changes in temperature between -57°C to +71°C following a 4-hour soak at each temperature.

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3.8.4 Altitude.

3.8.4.1 Operation. The commander's display shall not be damaged when operated at altitudes up to 10,000 feet above sea level.

3.8.4.2 Transportation. The commander's display shall not be damaged when subjected to altitudes up to 40,000 feet above sea level.

3.8.5 Shock.

3.8.5.1 Basic shock. The commander's display shall not be damaged when subjected to basic shock of 35 g's at 15 milliseconds duration.

3.8.5.2 High intensity shock. The commander's display shall not be damaged when subjected to high intensity shock of 100 g's at 4 milliseconds duration.

3.8.5.3 Bench handling shock. The commander's display shall not be damaged when subjected to shocks encountered in bench handling.

3.8.6 Vibration. The commander's display shall not be damaged when subjected to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three mutually perpendicular axes. (Group C: 5 minutes per axis.)

Random: 6×10^{-3} g^2/Hz, 5 to 500 Hz

Sine: 1.06 g's, X axis
1.51 g's, Y axis
1.06 g's, Z axis

Octave 1 42 to 84 Hz

Octave 2 84 to 168 Hz

3.8.7 Waterproofness. The commander's display shall withstand water hosing at a minimum of 5 pounds per square inch (psig) through a 1-inch nozzle parallel to the joints from a distance of 3 feet maximum for a time of 5 minutes minimum, without intake of water or damage.

3.8.8 Humidity. The commander's display shall operate and not be damaged by exposure to 100 percent relative humidity.

3.8.9 Dust. The commander's display shall not be damaged when exposed to dust at velocities up to 1750, ±250 feet per minute (fpm) for 12 hours.

3.9 Reliability. Not applicable.

3.10 Workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Materials handling. The supplier is responsible for processes, parts, and materials used in manufacturing the commander's display in accordance with referenced specifications, standards, and drawings as specified herein.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)
c. Inspection of packaging (4.10)

4.3 First article inspection. When specified in the contract, the contractor shall furnish 5 commander's displays for first article inspection (see 6.2).

4.3.1 Inspection. Each assembled first article commander's display shall be examined in accordance with table I. Presence of one or more defects shall be cause for rejection of the commander's display.

4.3.2 Preconditioning. Each commander's display which has passed the inspection of 4.3.1 shall be subjected to the preconditioning of 4.6.

TABLE I. Inspection.

Defects	Requirement paragraph	Inspection criteria
101. Any part or component missing or damaged	3.2	MIL-STD-252
102. Finishes not protecting substrates (scratches, etc)	3.2	MIL-STD-252
103. Scratched or chipped optics	3.2	MIL-STD-252
104. Improper or absent marking or labels	3.2	MIL-STD-252
105. Color of finishes incorrect	3.2	MIL-STD-252
106. Workmanship not as specified	3.10	MIL-STD-252
107. Contrast control	3.7.10.1	4.7.10
108. Polarity switch	3.7.10.2	4.7.10
109. Brightness control	3.7.10.3	4.7.10
110. Commander and gunner control	3.7.10.4	4.7.10

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4.3.3 Tests.

4.3.3.1 First article inspection. From commander's displays that have passed preconditioning of 4.3.2, five (5) first article commander's displays shall be subjected to all tests of table II, in any order. Unless otherwise specified, the tests of table II shall be performed with the commander's display at the ambient conditions (see 6.3). Failure of any commander's display to pass all tests in table II shall constitute failure of first article.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Luminance uniformity	3.7.1	4.7.1
Power	3.7.2	4.7.2
OTF	3.7.3	4.7.3
Apparent angular magnification	3.7.4	4.7.4
Spatial uniformity	3.7.5	4.7.5
Effective focal length	3.7.6	4.7.6
f/number	3.7.7	4.7.7
Gain	3.7.8	4.7.8
Resolution	3.7.9	4.7.9
Contrast control	3.7.10.1	4.7.10
Polarity switch	3.7.10.2	4.7.10
Brightness control	3.7.10.3	4.7.10
Commander and gunner control	3.7.10.4	4.7.10
Sealing	3.7.11	4.7.11
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Temperature shock	3.8.3	4.8.3
Altitude	3.8.4	4.8.4
Shock	3.8.5	4.8.5
Vibration	3.8.6	4.8.6
Waterproofness	3.8.7	4.8.7
Humidity	3.8.8	4.8.8
Dust	3.8.9	4.8.9

4.4 Quality conformance inspection.

4.4.1 Inspection. Each assembled commander's display shall be examined in accordance with table I. Presence of one or more defects shall constitute failure of that commander's display.

4.4.2 Preconditioning. Each commander's display which has passed the inspection of 4.4.1 shall be subjected to the preconditioning of 4.6.

4.4.3 Group A inspection. Each commander's display that has passed the preconditioning of 4.4.2 shall be inspected for the attributes specified in table III in any sequence. Failure of any commander's display to pass all the tests of table III shall constitute failure of that commander's display.

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4.4.4 Group B inspection. Group B inspection is not applicable.

4.4.5 Group C inspection. Inspection lots (see 6.2) shall be formed from commander's displays which have passed the Group A inspections of 4.4.3.

4.4.5.1 Monthly Group C. Five commander's displays shall randomly be selected from each lot and inspected for the attributes specified in table IV. Failure of any commander's display to pass all the tests of table IV shall constitute Group C failure.

4.4.5.2 Trimonthly Group C. Two commander's displays which have passed the tests of 4.4.3 and 4.4.5.1 shall randomly be selected from the first lot and every third lot thereafter. The two commander's displays shall be tested per table V in the order listed. Following completion of the tests of table V, the tests of table III and IV shall be reperformed. Failure of any commander's display to pass all tests of tables III, IV and V shall constitute Group C failure.

4.4.5.3 Group C failure criteria. Actions relative to Group C failure shall be as specified in the contract or purchase order (see 6.2).

4.4.5.4 Reinspection of Group C sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group C inspection may be accepted on contract or order provided all damage is repaired and the repaired units are subjected to and pass 4.4.3.

TABLE III. Group A inspection.

Inspection	Requirement Paragraph	Test Paragraph
Gain	3.7.8	4.7.8
Resolution	3.7.9	4.7.9
Contrast control	3.7.10.1	4.7.10
Polarity switch	3.7.10.2	4.7.10
Brightness control	3.7.10.3	4.7.10
Commander and gunner control	3.7.10.4	4.7.10

TABLE IV. Group C monthly inspection.

Inspection	Requirement Paragraph	Test Paragraph
Luminance uniformity	3.7.1	4.7.1
Power	3.7.2	4.7.2
Apparent angular magnification	3.7.4	4.7.4
Spatial uniformity	3.7.5	4.7.5
Effective focal length	3.7.6	4.7.6
f/number	3.7.7	4.7.7
Sealing	3.7.11	4.7.11
OTF	3.7.3	4.7.3

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TABLE V. Group C trimonthly inspection

Inspection	Unit	Requirement paragraph	Test paragraph
Low temperature	2	3.8.1	4.8.1
High temperature	2	3.8.2	4.8.2
Temperature shock	2	3.8.3	4.8.3
Altitude	1	3.8.4	4.8.4
Shock	1	3.8.5	4.8.5
Vibration	1	3.8.6	4.8.6
Waterproofness	1	3.8.7	4.8.7
Humidity	1	3.8.8	4.8.8

4.5 Group D inspection. Not applicable.

4.6 Preconditioning. Each commander's display shall be subjected to a minimum of 6 preconditioning cycles. Each cycle shall conform to the profile of figure 1. The temperature extremes shall be $-32^{\circ}C$ and $+52^{\circ}C$. The system shall be vibrated at 2.2 g's at any nonresonant frequency between 20 and 60 Hz for 10 minutes each operating hour. If damage occurs, the system shall be removed from this test, repaired, and returned to preconditioning. The preconditioning shall continue for at least one full cycle without additional failures after repair.

4.7 Tests. All tolerances herein are absolute. Any test measuring equipment tolerances shall be subtracted from the commander's display test measurement. Unless otherwise specified, no internal adjustments or components replacement shall be made on any commander's display subsequent to the start of any of the tests specified herein.

4.7.1 Luminance uniformity. The uniformity of the display shall be measured by using a photometer to scan the display in the vertical direction in the center, right and left half of the biocular with uniform stop illumination. Record the photometer output using an x-y recorder. Luminance variance in excess of ±15 percent from the average display luminance over the central 50 percent of the display shall constitute failure of this test.

4.7.2 Power. The commander's display shall be connected to a variable dc power source. A dc voltmeter shall be used to measure the input voltage at the input connector (3J1-C, HI and 3J1-D, LO). A dc ammeter shall be connected in series with the input line. The input voltage shall be adjusted to 5 Vdc and the input current measured. The product of the input voltage and input current shall not exceed 3.5 watts. Connect the power source to 3J1-U(HIGH) and 3J1-V(LOW). Adjust the input voltage to 5 vdc and measure the current. The product of the input voltage and the input current shall not exceed 1.5 watts. Measurement of input power between pins (3J1-C, HI and 3J1-D, LO in excess of 3.5 watts or between pins 3J1-U(HIGH) and 3J1-V(LOW) in excess of 1.5 watts shall constitute failure of this test.

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4.7.3 OTF. The commander's display shall be tested to ensure that it meets the requirements for optical transfer function. The system shall be tested for modulation transfer function (MTF). The system, operating in a linear region of the Signal Transfer Curve, shall be focussed on a target which is projected from a collimator. The target, either edge of slit, shall be placed in the center 25 percent (on-axis) of the display FOV. The image at the eyepiece of the target shall be scanned by a detector and the output used to compute the system optical transfer function. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Apparent angular magnification. The display shall be presented with a 4 bar, 7:1 aspect ratio, target so that the target appears on axis (centered in the display's eyepiece). A suitable telescope shall be used to image the target, at the biocular eyepiece, onto a scanning slit. Connect the scanning slit to a photometer and an x-y recorder. Scan the scanning slit across the bar target. Record the output of the photometer on the x-y recorder. Calculate magnification as follows:

$$\text{Magnification} = \frac{f_o}{f_{out}}$$

Where: f_o = target frequency

f_{out} = measured frequency at the output of the display eyepiece

Failure to meet the requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Spatial uniformity. Following the procedure of 4.7.4 for angular magnification, the spatial uniformity shall be measured by recording the angular magnification at four (4) points on the display other than the on axis point. These four (4) points shall be the center point of each of the four (4) quadrants of the display. The four bar targets shall be scanned horizontally in each quadrant utilizing the scanning slit. The distortion (uniformity) for each of the four (4) points shall be calculated as follows:

$$\text{Distortion} = \frac{(\text{MAG on axis} - \text{MAG})}{\text{MAG on axis}}$$

Measurement of distortion greater than $\pm$10 percent at any of the four points shall constitute failure of this test.

4.7.6 Effective focal length. The effective focal length of the relay system of the commander's display shall be measured using either Method 1 or Method 2 or the calibrated focal length method of MIL-STD-150 and shall comply with 3.7.6. Failure to meet the requirements of 3.7.6 shall constitute failure of this test.

4.7.7 f/number. A 11.1, $\pm$0.1 mm diameter circular aperture shall be located 67.6, $\pm$0.7 mm from the front of the housing of the commander's display optical relay system. The exit pupil of the system shall be viewed from a position at which the image intensifier is nominally located. The observer

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shall confirm that no vignetting or aperturing is occurring. Failure to observe the 11.1 mm diameter circular aperture shall constitute failure of this test.

4.7.8 Gain. Luminance gain is defined as a ratio of screen brightness (measured in foot-lamberts) to stop illumination (measured in foot-candles). A 2854 K, ±100 K tungsten filament lamp shall be adjusted to uniformly fill the stop of the commander's display and provide a photocathode illumination on the I^2 tube in the range of 5.0, $\pm 0.1 \times 10^{-2}$ foot-candles. The output of the I^2 tube is then measured and the ratio taken. Failure to meet the requirements of 3.7.8 shall constitute failure of this test.

4.7.9 Resolution. The resolution shall be tested according to MIL-STD-150 using an Air Force Resolution Target (100 to 1 contrast) and a filtered light source approximating the spectrum of red LED (λ= 600 nm). Failure to meet the requirement of 3.7.9 shall constitute failure of this test.

4.7.10 Controls. Make all the control adjustments indicated in table VI and record the results. All pin locations are on connector 3J1. All measurements are resistance measurements and are made between the pins indicated in table VI. Any reading which is not within the acceptable values shall constitute failure of that test.

TABLE VI. Control inspection

Control	Control Position	Location of Measurement	Acceptable Values (ohms)
Contrast	Full range	G&J	25K ± 5%
	Full clockwise (CW)	J&H	25K ± 5%
	Full counter-clockwise (CCW)	J&H	0
	CW	G&H	0
Polarity	Up	B&N	0
	Down	B&N	Infinity
Brightness	Full range	K&M	25K ± 5%
	CW	L&M	25K ± 5%
	CCW	L&M	0
	CW	K&L	0
	CCW	K&L	25K ± 5%
Connector interlock	---	E&A	0
Commander and gunner	Commander	E&F	0
	Gunner	E&F	Infinity

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4.7.11 Sealing. Before the start of the sealing test, all plates, covers, access doors, etc., shall be removed and installed two times. The sealing test shall be performed using a pressure gage capable of reading between 0.25 and 6.0 psig with a resolution of 0.2 psig or less. Pressurize the unit with dry nitrogen gas having a dew point at least as low as -32°C to 5.0, ±1.0 psig. Retain the applied pressure in the unit for a minimum of 2 hours. All knobs, switches, levers, actuators, and controls shall be moved to their extreme positions at least three times during the 2-hour period. At the end of the 2-hour period, there shall be no evidence of leakage in excess of 0.2 psig as indicated on the pressure gage. Upon satisfactory completion of the sealing test, the units shall be flushed with dry nitrogen gas specified herein and the unit shall be sealed with a delivery pressure as specified in 3.7.11. Failure to meet the requirements of 3.7.11 shall constitute failure of this test.

4.8 Environmental tests. Unless otherwise specified herein, environmental tests shall be performed in accordance with MIL-STD-810. The performance tests required after the completion of the individual tests of MIL-STD-810 shall be 4.7.8 and 4.7.9. When MIL-STD-810 requires a performance test during environmental test, the tests in table III shall be conducted.

4.8.1 Low temperature. The commander's display shall be tested in accordance with MIL-STD-810, Method 502, Procedure I, except that the storage period (Step 2) shall be 12 hours at -57°C, and the operating temperature (Step 4) shall be -32°C. Failure to meet the requirements of 3.8.1 shall constitute failure of this test.

4.8.2 High temperature. The commander's display shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The operating temperature (Step 7) shall be +52°C. Failure to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Temperature shock. The commander's display shall be tested in accordance with MIL-STD-810, Method 503.1, Procedure I. Failure to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Altitude. The commander's display shall be tested in accordance with MIL-STD-810, Method 500, Procedure I, except the chamber pressure, operating, shall be 523.0 mm Hg or 10,000 feet above sea level. The altitude, nonoperating, shall be 40,000 feet and shall be maintained for a period of not less than one hour, and shall be performed prior to the operating altitude test. Prior to the operating test, the commander's display shall be purged with dry nitrogen. Failure to meet the requirements of 3.8.4 shall constitute failure of this test.

4.8.5 Shock. The commander's display shall be tested in accordance with MIL-STD-810, Method 516.1, Procedures I, IV and V. Figure 516.1-2 shall apply to Procedures I and IV. The parameters for Procedure I shall be 35 g's peak value with 15 ms duration. The parameters for Procedure IV shall be 100 g's peak value with 4 ms duration applied by 3 shocks along each axis (18 shocks total). Failure to meet the requirements of 3.8.5 shall constitute failure of this test.

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4.8.6 Vibration. The commander's display shall be tested in accordance with MIL-STD-810, Method 514.1 and procedures as modified herein. Mount the commander's display on the vibration table with the commander's display non-operating, and vibrate it in each of three mutually perpendicular axes to the random and superimposed sinusoidal levels of 3.8.6 for 15 minutes. (Group C: 5 minutes per axis.) The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Waterproofness. Subject the commander's display to a water pressure of 5 psig through a 1-inch nozzle from a distance of 3 feet maximum in a direction parallel to the joints for 5 minutes. Wipe the system dry, remove the covers and inspect for water inside. Verify that the system is operable. Inability to operate following experiencing the water pressure test shall constitute failure of the test.

4.8.8 Humidity. The test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure II. Failure shall be as defined in MIL-STD-810.

4.8.9 Dust. The test shall be conducted in accordance with MIL-STD-810, Method 510, Procedure I. Failure shall be as defined in MIL-STD-810.

4.9 Reliability. Not applicable.

4.10 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Commander's Display, AN/VSG-2 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Commander's Display, AN/VSG-2 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Commander's Display, AN/VSG-2. Each Commander's Display, AN/VSG-2, shall be individually packed method III as follows: Place each unit within a barrier bag fabricated of material conforming to MIL-B-117, type I,

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class B, style 2. Closure may be by staples, tape, adhesive or heat seal. Cushion each unit on all surfaces with material conforming to MIL-P-26514, type I, class 1, designed to protect all projections and absorb the shock of impact in handling and transit. Place each cushioned item into a close-fitting box conforming to PPP-B-601, overseas type. Place the contents, directly under the cover of the box. To assure reusability of the box and to provide easy removal of the cover, fasteners-container closure conforming to MS-24535 shall be utilized. Strapping shall be in accordance with the appendix of the box specification except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188 (see 6.2).

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Each Commander's Display, AN/VSG-2, unit packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container.

5.2.1.1 Unitization. Palletization shall be required when: Containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. Each Commander's Display, AN/VSG-2, unit packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container, except that the box shall be domestic type.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Commander's Displays, AN/VSG-2, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or W3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188 (see 6.2).

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5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129 and as follows:

"REUSABLE PACKAGE
DO NOT DESTROY
DO NOT OPEN PRIOR TO USE"

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The commander's display is intended for use in the AN/VSG-2.

6.2 Ordering data. Procurement documents will specify, as a minimum, the following:

a. Title, number, and date of this specification.
b. That quality conformance inspection lots for Group C testing (see 4.4.5) shall correspond to the quantity of commander's displays to be delivered each month.
c. If there is to be a requirement to perform first article test, a quantity of five (5) first article commander's displays are required (see 4.3).
d. Corrective action requirements for Group C inspection (see 4.4.5.4).
e. Disposition of first article commander's displays. First article commander's displays shall be returned to satisfactory operation and submitted for inspection and acceptance as part of a quality conformance commander's display lot.
f. Level A, level B or commercial preservation and packing (see Section 5).
g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
h. If the commander's display is to be subjected to environmental testing as part of the AN/VSG-2 system level test at a sampling rate equal to or greater than that contained herein, then the environmental tests of this specification may be waived.
i. MIL-STD-810B shall be used for environmental tests of 4.8.

6.3 Definitions. For the purpose of this specification, the following definitions apply.

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6.3.1 Damage. Damage is defined as degradation in specified performance, breakage, loosening, shift, failure of any finish, hardware, connection or component, evidence of condensation or moisture within the system, and deterioration in quality, or failure to operate in accordance with the requirements of this specification.

6.3.2 "g". g is defined as an acceleration or deceleration of 9.8 meters per second per second.

6.3.3 Ambient conditions. Ambient conditions are defined as follows:

Temperature	23°C, ±10°C (73°F, ±18°F)
Relative humidity	50 percent, ±30 percent
Atmospheric pressure	725, +50 mm of mercury -75 mm of mercury (28.5, +2.0 inches of mercury -3.0 inches of mercury)

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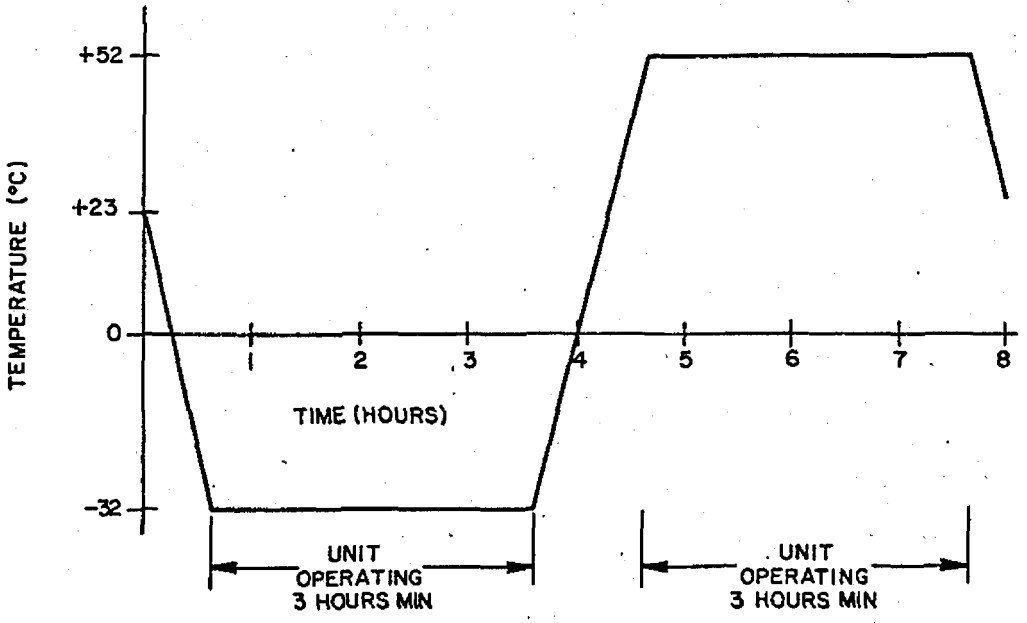


FIGURE 1. ONE PRE-CONDITIONING CYCLE

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-49380 (ER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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